



The influence of artificial intelligence chatbot problem solving on customers' continued usage intention in e-commerce platforms: an expectation-confirmation model approach

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ABSTRACT

The rapid advancement of artificial intelligence (AI) technology has led to its widespread application across various fields, including e-commerce platforms. AI chatbots, powered by large data models, have emerged as a crucial tool for enhancing customer service and user experience. This study aims to investigate the impact of AI chatbot problem-solving capabilities on users' intention to continue using these services within e-commerce platforms. By integrating elements from the Expectation-Confirmation Model (ECM) and the Technology Acceptance Model (TAM), a comprehensive research model is proposed to examine the relationships between problem-solving, confirmation, perceived ease of use, satisfaction, trust, and continued usage intention. The study employs structural equation modelling (SEM) to analyse data collected from 315 participants through an online questionnaire. The findings reveal that problem-solving abilities positively influence users' confirmation, which in turn positively affects satisfaction, perceived ease of use, and trust. Furthermore, perceived ease of use exhibits a positive impact on satisfaction, trust, and continued usage intention. Satisfaction and trust have a positive impact on continued usage intention, with satisfaction showing a stronger influence than both problem-solving ability and ease of use. The study contributes to the theoretical understanding of user perceptions and behaviours in AI-driven interactions within e-commerce platforms. It highlights the importance of problem-solving capabilities, perceived ease of use, satisfaction, and trust in driving users' continued engagement with AI chatbots. Practical implications for AI technology developers and e-commerce companies are discussed, emphasizing the need to focus on enhancing chatbots' problem-solving proficiency and user-friendliness to foster long-term user retention. Future research directions are proposed, addressing the study's limitations and exploring the generalizability of the findings beyond the e-commerce context.

1. Introduction

In recent years, the development of artificial intelligence (AI) technology has accelerated rapidly, leading to its swift adoption across various fields. Generative AI systems are becoming instrumental in business expansion in many areas. Generative artificial intelligence (AI) which refers to algorithms such as ChatGPT has evolved significantly and has not only transformed human – computer interaction but also has the capacity to create diverse content, including audio, code, text, etc. (Akhtar, 2024). Tasks that once required multiple workers can now be easily completed by AI systems, such as facial recognition systems (Gupta et al., 2023), cloud data analysis, and translation of languages (Li

et al., 2024). With the dynamic changes in AI technology, enterprises are increasingly deploying AI technology in business-related services to enable them engage effectively with customers and optimise value creation. For example, The North Face (clothing company) uses AI technology to help customers choose clothes that fit better (Gursoy et al., 2019). Uniqlo developed a shopping assistant (Uniqlo IQ) to help consumers search for information about items online and answer relevant questions (Yuan et al., 2023). Applications based on Large Language Models (LLMs) can provide personalized recommendations to relevant clients across a wide range of industries (Kolasani, 2023). The Chinese e-commerce platform's AI chatbot application based on the LLMs can speedily and accurately respond to the questions raised by customers

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and offer useful guide, thus enhancing the interaction between customers and the e-commerce platform (Ngo Tran et al., 2024).

Notably, the e-commerce platform can enable AI communication capability towards achieving effective communication with customers and thus improving the customer service effect and efficiency of the platform (Asadi and Hemadi, 2018). Although AI has been widely used in e-commerce platform services and marketing strategies, user experience of AI customer service, which is one of the key areas of interest of many e-commerce companies and platforms, is relatively under-explored (Følstad and Skjuve, 2019; Sonntag et al., 2022). Most relevant studies in this field focus mainly on product classification and recommendation of AI technology in customer service (Marwade et al., 2017; Shafi et al., 2020). This is a critical research space in today's post-pandemic era, where technological evolution is increasingly shaping how we do things, and in particular buy (or not buy) consumer decisions. The notion of 'customer is king' is gaining increasing significance, due to intense competition and technology enabling impact on consumers (Opote et al., 2025). Therefore, achieving the 'cutting edge' would require that marketers understand and respond effectively to customers. AI customer service technology plays a crucial role in enhancing the user experience by addressing issues consumers encounter while shopping online and resolving them promptly, thus improving the interaction between e-commerce platforms and customers (Prentice & Nguyen, 2020; Sonntag et al., 2022). Additionally, AI-based applications can reduce workloads, thereby increasing platform efficiency and lowering operational and administrative costs (Cobos et al., 2016; Mydyti & Kadriu, 2021). On their part, Hsiao & Chen, (2022) emphasise that due to the versatility of AI systems, chatbots powered by large models can offer 24/7 online customer support. The challenge therefore is ensuring that AI chatbot customer service meets or exceeds the service quality of human agents on traditional e-commerce platforms.

Heeding the advocacy in recent literature (Følstad et al., 2021; Hollebeek et al., 2024), a central focus in this study is to understand how consumers perceive the design and content of AI chatbots and how that ultimately influences their continued usage intention and related aspects. In terms of functionality, AI chatbots are focused on improving

problem-solving capabilities (Ng & Lin, 2022) and enhancing their ability to deliver services that meet user needs more effectively (Liu et al., 2023). For instance, during interactions, consumers can ask about product details, shipping information, pricing, and more. The chatbot will then provide appropriate solutions, as illustrated in Fig. 1. It is important to underline that customer service chatbots as well personal assistant chatbots were considered in this study. Given the foregoing discourse, the present study aimed to understand research question (RQ1):

RQ1: What is the impact of AI chatbot customer service's problem-solving capability on users' sustained usage intention in e-commerce platforms?

In this study, we integrated elements of the Expectation-Confirmation Model (ECM), problem-solving factors, DeLone and McLeans IS success model and the Technology Acceptance Model (TAM) towards understanding perceived ease of use and users' intentions for continued usage. This integration of theories draws from prior literature that rationalises these models as suitable frameworks for examining the continued use of e-learning platforms: ECM (Lee, 2010; Alnaser et al., 2023), TAM (Brahim et al., 2018; Sonntag et al., 2022; Almogren and Aljammaz, 2022) and DeLone and McLean IS Success model (DeLone and McLean, 2003; Mamun et al., 2020; Mishra et al., 2023). The explored model employs three key variables—confirmation, satisfaction, and intention to continue using—to understand user behavioural intentions. Users' experiences with technology can also be influenced by additional factors, such as subjective behaviours and recommendations from others (Ngo and Le, 2023). Research conducted by Rather et al. (2021) further indicates that users' attitudes significantly influence their behavioural intentions, with trust and satisfaction serving as critical drivers for continued technology usage. Moreover, factors such as performance expectancy, effort expectancy, social influence, hedonic motivation, inconvenience, anthropomorphism, and attitudes toward technology can positively impact users' ongoing engagement with chatbots (Melián-González et al., 2021).

In another perspective that adds to the discourse on users' intentions and related factors, DeLone and McLean IS Success model (2003) builds on the communication theory-based perspectives of Shannon and



Fig. 1. Screenshot examples of AI chatbots from different e-commerce platforms (Li-Ning store on Taobao platform, Adidas store on Jingdong platform).

Weaver (1949) and Mason (1978) to theorise IS Success model that includes systems quality, information quality, and “use, user satisfaction, individual impacts,” and “organizational impacts.”

An individual's intention to use a system is shaped by his/her attitude toward its use and the perceptions of its ease of use (Saadé and Kira, 2009). Furthermore, users' engagement with e-commerce platforms is characterized by long-term and adaptive behaviours (Singh & Chakrabarti, 2023). A comprehensive approach is essential towards providing a more nuanced understanding of the factors influencing users' intentions to continue using technology. As the system becomes easier to use, users perceived self-efficacy regarding their ability to comfortably engage with the system increases. Therefore, this study addresses the following research question:

RQ2: What is the relationship between perceived ease, based on problem-solving, and consumers' intention to sustain usage in e-commerce platforms?

Trust in a technology is also a critical factor influencing user's adoption of that technology, particularly in the online setting (Aslam et al., 2020). Trust is an intangible concept (Pala, 2024) that can enhance customer purchase intention (Nazir & Tian, 2022), drive customer retention (Zhang et al., 2022), and also regulate emotions to impact sales volume (Wang et al., 2023). Thus, this study addresses the following question:

RQ3: How does consumers trust influence their continued usage intention of AI chatbot customer service in e-commerce platforms?

This study explored the impact of AI chatbot customer service on the ability to drive customer problem solving. This research focus would enable better understanding of customer opinions on e-commerce platform type AI chatbots. That understanding can help the program developers of e-commerce companies improve the service quality and content of their AI chatbots, improve customer satisfaction with AI customer service in the e-commerce platform, and enhance the problem-solving ability, interaction ability and user experience of AI customer service, and extendedly increase users continued use of AI chatbots. Furthermore, the results of this study will offer references for future research and provide guidelines for e-commerce platforms to design more comprehensive AI chatbots for optimising value creation.

2. Literature review and hypothesis

2.1. Defining the concepts

2.1.1. AI chatbot in e-commerce platform

In recent years, the development of AI-based chatbots on e-commerce platforms has emerged as a viable alternative to traditional human customer service (Wang et al., 2022; Song et al., 2022). The deployment of AI chatbots in e-commerce is also a highly profitable venture. According to Juniper Research (2023), global spending on chatbots in retailing is projected to skyrocket from \$12 billion in 2023 to nearly \$72 billion by 2028. A significant 83 % of consumers worldwide require assistance during online shopping (Li and Wang, 2023), typically relying on human employees. However, advancements in AI technology have significantly enhanced customer service efficiency through AI-powered chatbots. These chatbots offer 24/7 support, handling tasks like answering frequently asked questions (FAQs), providing product recommendations, and resolving basic issues. Current AI chatbots can address 90 % of consumer inquiries in real time, thereby improving operational efficiency while allowing human employees to focus on more complex tasks, ultimately enhancing service quality (Kumar et al., 2021).

As AI chatbots become integral to e-commerce platforms, they serve as essential tools that complement human employees, taking on an increasing share of customer service tasks (Ruan and Mezei, 2022). The widespread adoption of AI chatbots by major brands like Nike and Adidas has generated significant academic interest in their application within the e-commerce sector. For example, research by Chen et al.

(2021) found that chatbot responsiveness and availability significantly boost customer satisfaction. Similarly, the perceived empathy and friendliness of e-commerce chatbots can positively influence consumer trust and subsequent purchase behaviours (Cheng et al., 2022). Additionally, Chattaraman et al. (2024) argue that the cognitive processing ability and product knowledge of AI chatbots play a crucial role in shaping consumers' shopping experiences and behavioural intentions.

While the literature on chatbots continues to grow, consumers initially tend to have low expectations of chatbot services (Lou et al., 2022; Song et al., 2022). However, increasing numbers of users are willing to engage with AI chatbots. According to Li et al. (2020), only 7.6 % of consumers reported using AI chatbots during online shopping, and this percentage continues to decline. Moreover, in the pragmatic aspect of the survey, 84.8 % of users expressed that they would accept any form of customer service, as long as it solves their problems (Li et al., 2020). Despite increasing research, there is still limited understanding of how to effectively leverage AI chatbots' problem-solving abilities to foster consumers' willingness to continue using them on e-commerce platforms.

The core value of chatbots lies in their ability to effectively resolve customer issues. Over time, consumers develop an understanding of a chatbot's strengths and weaknesses through repeated interactions. As a result, assessing users' continued usage intentions is more critical than measuring their initial adoption decisions, as the former reflects their long-term experiences and satisfaction (Bhattacharjee, 2001; Lee and Kwon, 2011). Although previous studies have examined chatbot acceptance and initial adoption (Kasilingam, 2020; Rese et al., 2020; Sheehan et al., 2020), relatively few have focused on the factors driving continued usage. Most literature has explored the reasons behind continued intentions in chatbot services (Ashfaq et al., 2020; Cheng and Jiang, 2020).

This study extends existing research by focusing on the role of problem-solving capabilities in AI chatbot design and their impact on consumer satisfaction, trust, perceived ease of use, and continued usage intentions. Understanding this aspect is crucial, as sustained use is a key determinant of the ultimate success of AI chatbot deployments (Myin, 2024). Emergent insights can guide e-commerce platforms in optimizing their chatbots' problem-solving functionalities, thereby maximizing the potential of AI-powered customer service.

2.1.2. Expectation-Confirmation model (ECM)

In the social science research field, Expectation confirmation theory (ECM) is one of the well-known theories that is considered to evaluate and measure whether users can continue to use technology. Developed in 1980, the ECM model focuses on evaluating a user's continue usage intention, confirmation, and satisfaction as shown in Fig. 2. ECM can be used to explain a customer's willingness to re-use an AI chatbot customer service in an e-commerce platform. The development of ECM is based on the self-perception theory (Bem, 1972). Bhattacharjee (2001) believes that the causal relationship between technology and service is adopted to analyse and explain users' willingness to continue to use information technology and corresponding services. This approach provides a theoretical framework for examining the interplay

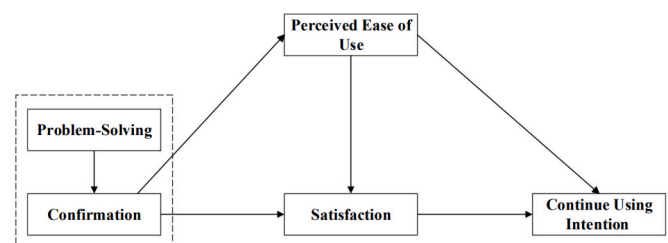


Fig. 2. Based on problem-solving expectation-confirmation model (Alnaser et al., 2023).

between expectations and concepts. Therefore, ECM has been developed to interpret the customer's expectations when using and receiving feedback from others or themselves using relevant technologies (Mamun, 2020). The customer determines the level of post-acceptance behaviour through continuous use and feedback (Mishra et al., 2023).

Previous studies have mainly used ECM as a framework to promote a more comprehensive understanding of research model methodologies (Wandira et al., 2024). In this paper, based on user-centred theory, an AI chatbot customer service is introduced to address problems in users' experience. Factors like perceived usefulness, perceived ease of use, satisfaction, and trust were used as key determinants in ECM. Therefore, this study integrates two models – ECM and Technology Acceptance Model (TAM), drawing also from relevant theoretical substances from the D & M IS Success model (DeLone and McLean, 2003) to design a framework to explain effective ways for users to solve problems through AI customer service and continue usage of the platform. Confirmation and satisfaction are important factors in ECM for measuring users' continued usage. As Hsu and Lin (2015, p. 47) argue, confirmation is “the users' assessment of the congruence between expectation and actual performance of using an information system or service.” Confirmation plays a key role in users' perceived satisfaction with systems and services, while satisfaction relates to users' intention to continue using platform systems and services. User satisfaction is often defined as “the extent to which users believe the information system meets their information requirements; it is also referred to as user information satisfaction” (Liu et al., 2020, p. 2412).

2.1.3. Technology acceptance model (TAM)

Technology acceptance model (TAM) is a theory expanded and developed based on Theory of Reasoned Action (TRA) theory, which can explain user acceptance in a more detailed way (Davis, 1989). TAM puts forward two specific measurement indicators, namely perceived usefulness, and perceived ease of use, which are the main factors in measuring user acceptance. Perceived usefulness is defined as:

“the degree to which a person believes that using a particular system would enhance his/her job performance, and perceived ease of use is defined as the degree to which a person believes that using a particular system would be free of physical and mental effort” (Davis, 1989, p. 320).

In addition, both perceived usefulness and perceived ease of use will affect consumers' attitude towards the use of the system, and similar to TRA, feelings and attitudes after use can determine consumers' behavioural intentions (Tae et al., 2024). TAM has been widely used to measure consumer acceptance of various types of technology, including e-commerce platforms, email, word processing systems, etc. (Owusu et al., 2022; Yu and Huang, 2020).

The TAM incorporates external variables such as individual differences, situational constraints, organizational characteristics, and system attributes, all of which collectively influence behavioural outcomes. For example, Zarouali et al. (2018) proposed the Consumer Acceptance Technology (CAT) model to study the effectiveness of chatbots for brands on Facebook, incorporating factors like perceived usefulness, ease of use, and helpfulness. Van den Broeck et al. (2019) examined how the perceived helpfulness and usefulness of chatbots on Facebook influence the perceived intrusiveness of chatbot-initiated advertisements. However, the traditional TAM emphasizes the role of external factors in shaping individuals' internal decision-making processes regarding system adoption within organizational environments (Davis and Granić, 2024). These external variables exert both direct and indirect effects on perceived usefulness (PU), mainly through perceived ease of use (PEU), with immediate perceptions of utility being more influential than long-term considerations (Berbar, 2023). In contrast, Meyer-Waarden et al. (2021) argue that within the TAM framework, reliability and perceived usefulness are the most critical factors driving users' intent to reuse chatbots.

Furthermore, factors such as system interactions with users, direct user experiences, and system characteristics, along with domain knowledge, determine the user's perception of a system's ease of use (Cheong et al., 2021; Widjaja & Widjaja, 2022; Elshan et al., 2022). Previous research indicates that efficacy, intrinsic motivation, and cognitive absorption are determinants of PEU and user intention in consumer settings (Hsu, 2024; Butt et al., 2022). Given that AI chatbot technologies are fundamentally designed to address users' needs, PEU is an indispensable requirement. Considerable efforts have been devoted to crafting user-friendly graphical user interfaces in application development, recognizing the pivotal role of PEU (Cheah et al., 2021). Thus, the framework shown in Fig. 2 combines the single TAM factor (PEU) along with ECM features to measure customer using experience with AI chatbot in e-commerce platforms. This framework aims to evaluate the user experience regarding the adoption of AI chatbots within e-commerce platforms.

2.1.4. DeLone and McLean (D&M) IS success model

A further model that has featured in the discourse about users' intention, satisfaction and related factors is the D&M (2003) IS Success model. The D&S IS Success model builds off Shannon and Weaver's (1949) communication theory, Mason's (1978) information influence theory, and empirical insights from management information systems (MIS) studies from 1981-1987. Shannon and Weaver (1949) distil three communication level typologies. Within that theory, the *technical level* of communications implies the accuracy and efficiency of the communication system that produces information, while the *semantic level* connotes the success of the information in effectively conveying the meaning intended. On the other hand, the *effectiveness level* signifies the information effect on the receiver. Drawing on communication theory, Mason (1978) forwards a framework for measuring information system output that involves four output levels: *technical level output*, *semantic level output*, *functional output* and *influence or pragmatic level output*.

The D&S IS Success model is driven by the process understanding of IS and impacts, and the process is grounded on three components: “the creation of a system, the use of the system, and the consequences of this system use” (DeLone and McLean, 2003, p.16). This success model suggests an interrelation between the six conceptualised dimensions (system quality, information quality, system use, user satisfaction, individual impact[s] and organizational impact), leading to a success model that pictures same direction between causality and the information process. In this D&M IS Success model, ““systems quality” measures technical success; “information quality” measures semantic success; and “use, user satisfaction, individual impacts,” and “organizational impacts” measure effectiveness success.” (DeLone and McLean, 2003, p. 10). In this study, however, these IS Success model features are not per se included as constructs in the conceptualised framework for this study but the inherent theoretical substances are harmonised with the relevant aspects of the theoretical narratives that ground the hypotheses for the variables showcased in Fig. 3.

In exploring AI Chatbot problem solving and customers' continued usage intention and related elements, we build on these streams of theories (ECM, TAM and D&M IS model) given their conceptual affinity: they address in respective conceptualisations core service quality aspects related to customers' continued usage intention, such as system quality, information quality, use, user satisfaction, and usefulness. Next, the underlying hypotheses for Fig. 3 are explained.

2.2. Problem-solving ability

AI chatbot customer service is an information system that can be applied across various fields, including social networks, information dissemination, entertainment, and transportation, leveraging AI's big data models to assist users in completing tasks (Hsiao and Chen, 2022). The problem-solving capability of a system can significantly influence the relationship between system quality and user satisfaction (Gürkut

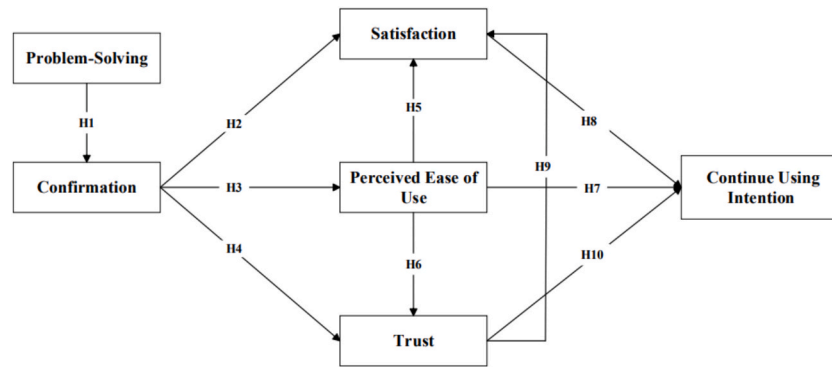


Fig. 3. The research model.

and Nat, 2017). Moreover, system quality directly impacts factors such as ease of use, functionality, and reliability (DeLone and McLean, 2003). Therefore, AI chatbot customer service has the potential to replace traditional human customer service (Ngai et al., 2021). Compared to other application systems, AI-powered chatbot services excel at resolving user issues and providing effective solutions. Additionally, they can offer personalized purchase recommendations, helping customers order products and fulfilling various user needs (Ngai et al., 2021). This makes AI chatbot customer service a versatile and efficient alternative to traditional methods, enhancing the overall user experience.

Problem solving is the ability of AI chatbot customer service to accurately understand and solve each customer's problem immediately. Sadhu et al (2024) believe that the main purpose of using AI customer service is to help improve work efficiency and productivity (given information quality benefits – DeLone & McLean, 2003), among which the convenience and ease of use brought by AI customer service are the main factors. On their part, Rane et al (2023) add that helping customers solve problems quickly can improve customer service, reduce related costs and enhance customer loyalty. Most companies need to create more loyal customers through services that help them solve problems easily (Rane et al, 2023). The AI chatbot customer service used by the users in this study will be highly valued for its ability to understand the problem and provide a quick solution. The ability to solve problems can enhance consumers' interest in online shopping (Fu et al., 2020). Users are confirming the ability of AI chatbot customer service to solve problems; therefore, the following hypothesis is proposed.

H1. The problem-solving ability of AI chatbots is positively correlated with customer confirmation of expectation of the e-commerce platform customer service.

2.3. Confirmation

User satisfaction with platform systems and services in the e-commerce industry is a key factor for users to continue to use the technology, and usually depends on the user's level of use in the actual use of platform systems to confirm (Pang et al., 2020). This foundation provides a good measure for user recognition of a technology, which represents the user's satisfaction and feelings after using a technology and the resulting impact (Rahi et al., 2021).

According to Festinger's (1962) theory of cognitive dissonance, when the user's actual after use perception reflects dissatisfaction, the user may suffer from psychological disharmony caused by disappointment. In order to mitigate this difference in perception that leads to disappointment before and after use, the feeling of confirmation would need to be improved (Hsieh, 2020). In addition, according to TAM's theory, the combination of the product itself, related services and users' actual feelings during use can significantly affect the perceptual and utilitarian factors of users' use (Cheah et al., 2023; Almogren & Aljammaz, 2022; Aldammagh et al., 2021; Ali, 2023). Previous studies

have adopted the concept of validation as a determinant of perceived usefulness and perceived ease of use (Alamsyah et al., 2022; Park E, 2020). For example, based on the data of 1380 respondents, Park (2020) found that users' confirmation of smart wearable devices plays a significant role in perceived ease of use, usefulness, and enjoyment. Similarly, validation of data from 363 respondents demonstrated that perceived usefulness, attitude, concentration, and perceived behavioural control significantly determined their confirmation. Applying that foundation to users who use AI customer service for problem solving on e-commerce platforms, this study proposes the below hypotheses.

H2. Users' confirmation of expectations is positively related to their satisfaction that the AI Chatbot has solved their problem.

H3. Users' confirmation of expectation is positively related to the perceived ease of use of the AI chatbot.

H4. Users' confirmation of expectations is positively related to their trust of the accuracy of the response from the AI Chatbot.

2.4. Perceived ease of use

In terms of AI customer service on e-commerce platforms, prior research defines perceived ease of use as each user's perception that AI customer service interaction does not require significant efforts to understand the problem of AI customer service and can easily solve the problem (Davis, 1989; Davis et al, 1989). More current research demonstrates that perceived ease of use is an important factor that can positively influence consumers after using systems and services (Wilson et al, 2021). The enhanced updated D&M model offers a clarification of the 'use' construct that demonstrates the importance of the construct (DeLone & McLean, 2003). They explain that "Use must precede 'user satisfaction' in a process sense, but positive experience with 'use' will lead to greater 'user satisfaction' in a causal sense" (cited in Petter et al., 2008, p.238). Further, they forward that increased user satisfaction will intensify intention to use, which also subsequently affects use. In the context of AI customer service on e-commerce platforms, perceived ease of use is positively correlated with users' satisfaction with shopping on mobile applications (Lopes et al, 2024). In addition, studies on medical and payment services (Kamal et al., 2020; Askari et al., 2020) have validated that relevance. Grounded on domain discourse, this study forwards the below hypotheses about AI chatbot customer service.

H5. Perceived ease of use is positively correlated with satisfaction that the AI chatbot has solved their problem.

H6. Perceived ease of use is positively correlated with trust of the accuracy of the response from the AI chatbot.

H7. Perceived ease of use is positively correlated with the willingness to continue using AI Chatbot.

2.5. Satisfaction

Consumer perceived satisfaction is a psychological construct that reflects users' emotions following their usage and purchase experiences (Bhattacharjee and Premkumar, 2004). This research specifically examines users' experiences and feelings after interacting with AI customer service on e-commerce platforms, emphasizing their perceptions of problem-solving capabilities and overall service use. According to the Expectation-Confirmation Model (ECM), consumers' intention to continue using a system or service is influenced by their satisfaction during the initial interaction. Sonntag, Mehmman, and Tuetebe (2022) explored the role of artificial intelligence in shaping customer satisfaction within the TAM-2 framework. Rajeh et al. (2021) found that user satisfaction in e-learning environments significantly affects their willingness to continue using the service. Furthermore, Kvale et al. (2020) argue that customer satisfaction is dependent on the AI agents' effectiveness in resolving issues faced during customer interactions. Based on these insights into the relationship between users and AI customer service on e-commerce platforms, the following hypothesis is proposed in this study:

H8. Users' satisfaction that the AI chatbot has solved their problem is positively correlated with their willingness to continue using AI chatbot.

2.6. Trust

Trust refers to a consumer's commitment to the reliability and functionality of a service (Przegalinska et al., 2019). Unlike interpersonal trust, some consumers are more inclined to trust systems, viewing them as more objective and rational than human interactions (Przegalinska et al., 2019). Trust remains a pivotal topic across various research fields. For instance, Beldad et al. (2010) studied the key factors that shape consumer trust in electronic services, particularly in the context of e-commerce. Hildebrand and Bergner (2021) believe that users' emotional trust in chatbots will influence their perceptions of the company. Cheng and Lin (2023) explored the link between community commerce and trust, demonstrating how trust drives purchasing behaviour while Manzoor et al. (2020) established that trust positively impacts businesses on social networking sites. Fatmawati and Fauzan (2021) found that word-of-mouth can enhance trust levels and Skripnuk et al. (2021) suggests that trust crises can affect sales volumes in the context of food safety risks. Przegalinska et al. (2019) identified crucial factors influencing user interactions with chatbots, while Nguyen, Chiu, and Le (2021) found that in the banking sector, trust surpasses customer satisfaction and perceived usefulness as the primary driver of continued chatbot usage. Further research (Cheng et al., 2021) argues that trust developed through users' experiences with chatbots on e-commerce platforms significantly affects their intention to continue using the technology. In summary, trust is a critical factor influencing users' willingness to engage with AI customer service on e-commerce platforms. Therefore, this paper proposes the following hypotheses:

H9. Users' trust in the AI chatbot service is positively correlated with satisfaction that the AI Chatbot has solved their problem.

H10 Users' trust in AI chatbot service is positively correlated with their willingness to continue using AI Chatbot

To sum up, this study incorporates multiple theories into the research model, which is necessary and valuable for measuring users' perception of AI chatbot customer service on e-commerce platforms. Based on the proposed hypotheses, the research model is shown in Fig. 3.

3. Research design and methodology

3.1. Questionnaire design

This research follows a two-part guideline of the questionnaire to test the designed theoretical model. The first part is about the demographics of the participants and the second part is about measuring structural

issues in the research model. In the first part, there are six questions about the basic information of the participants, including age, educational background, frequency of use, etc. The second part has 21 questions from seven constructs and each question was measured using the seven-point Likert scale, ranging from (1) Strongly Disagree to (7) Strongly Agree. All items were adapted from previous existing literature. The problem-solving items were adapted from Brandtzaeg and Følstad (2017), while the confirmation items were adapted from Oghuma et al (2016). The perceived ease to use were adapted from Briz-Ponce et al (2017). Furthermore, the perceived usefulness items were adapted from Park and Kim (2014), while the satisfaction items were adapted from Tan and Kim (2015). To measure trust, items were adapted from Przegalinska et al (2019), while continue use intention was measured using items from Davis et al (1989). The adapted measurement items are displayed in Table 1.

3.2. Data collection and sample size

The questionnaire was tested using convenience sampling, a non-random sampling method commonly used for audience selection in research. The researchers collected the questionnaires using a large data-gathering platform called Julestar (<https://www.wjx.com>). This platform is currently the largest questionnaire collection platform in China with nearly 99.52 million users and 12.12 billion questionnaires collected (Yuan et al, 2023). The data collection was conducted from June 30 to November 11, 2023. From the 500 distributed

Table 1
Questionnaire items.

Constructs	Items & sources
Problem Solving	(Brandtzaeg and Følstad, 2017) PS1: I think e-commerce AI chatbot can handle my problems when I use it. PS2: I think e-commerce AI chatbot can answer my question. PS3: I think e-commerce AI chatbot can understand the information that I need.
Confirmation	(Oghuma et al., 2016; Susanto et al., 2016) CF1: My experience with AI chatbot in e-commerce is better than what I expected. CF2: The majority of my expectations from using AI chatbot in e-commerce are confirmed. CF3: The functionalities of AI chatbot in e-commerce are better than what I expected.
Perceived Ease to Use	(Briz-Ponce et al., 2017; Munoz-Leiva et al., 2017) PEU1: My interaction with the AI chatbot in e-commerce has been clear and comprehensible. PEU2: Using AI chatbot in e-commerce is easy for me. PEU3: Interacting with the AI chatbot in e-commerce requires no mental effort.
Satisfaction	(Park and Kim, 2014; Tan and Kim, 2015) SF1: Overall, I am satisfied with AI chatbot in e-commerce. SF2: The AI chatbot in e-commerce I am currently using meet my expectations. SF3: I am pleased with my experience with using AI chatbot in e-commerce.
Trust	(Przegalinska et al., 2019; Oliveira et al, 2017) TR1: I consider the AI chatbot in e-commerce to be trustworthy. TR2: I consider that the AI chatbot in e-commerce can provide a reliable service. TR3: I consider that AI chatbot in e-commerce can provide a correct information.
Continue Using Intension	(Davis et al., 1989; Park and Kim, 2014; Mouakket, 2015) CUI1: After I used an AI chatbot in e-commerce, I am likely to continue to use AI chatbot in e-commerce. CUI2: I intend to consistently use AI chatbot in e-commerce as much as possible. CUI3: I intend to continue using AI chatbot in e-commerce rather than discontinue using them.

questionnaires, a total of 421 responses was received. In the data cleaning process, we found 9 blank responses, 31 incomplete answers, 58 abnormal and repetitive responses, and 8 instances of incorrect personal information (where age and education did not match). Consequently, 315 valid questionnaires were included in the final statistical analysis and the participants demographics are shown in Table 2. This sample size is considered sufficient for confirmatory factor analysis (CFA) and structural equation modelling (SEM) (Deb and Lomo-David, 2014). For the quantitative analysis, SPSS and AMOS were used. The first step in the quantitative analysis process involved conducting confirmatory factor analysis, followed by structural model analysis to validate the research hypotheses.

In this study, we adopted the approach proposed by MacCallum et al. (1996) to determine the statistical power and sample size using the Root Mean Square Error of Approximation (RMSEA). The analysis was conducted using a structural equation modeling (SEM) approach involving six latent variables (constructs), each measured by three observed variables. Based on the formula for calculating degrees of freedom $df = 2k(k+1) - t$, where k represents the total number of observed variables and t denotes the number of parameters to be estimated in the model, we calculated $k = 18$ and $t = 46$, resulting in $df = 125$.

Utilizing the semTools package, we conducted an RMSEA analysis, setting the initial RMSEA value at 0.10 (rmsea0) and the target RMSEA value at 0.05 (rmseaA). Given the degrees of freedom $df = 125$, a statistical power of power = 0.8, and a significance level of $\alpha = 0.05$, the required sample size was determined to be 56. Further analysis indicated that when the RMSEA value is set at 0.08, the standard sample size is 171. It is important to note that an RMSEA value of about 0.1 is generally considered to indicate poor model fit, while an RMSEA value between 0.05 and 0.08 is considered to represent a good fit. Additionally, the power analysis requires a power value greater than 0.8 to ensure sufficient statistical power.

Considering these factors, the actual sample size used in this study (315) exceeds the sample sizes calculated based on different RMSEA criteria (56 and 171). Therefore, it is legitimate to conclude that the sample size for this study satisfactorily meets statistical power requirements and ensures accuracy of model fit. This indicates that our research design is reasonable in terms of sample size, sufficient to support the exploration and analysis of relationships among latent variables. This statistical power sample-based conclusion is further supported by sample size benchmark for SEM analysis: sample size of 300 (Tabachnick and Fidell, 2013) and ration of observations (participants) to estimated parameters (variables) of 10 to 1 (Schreiber et al., 2006).

Table 2
Survey respondents.

Measure		Freq.	%
Gender	Male	118	37.46 %
	Female	197	62.54 %
Age	Under 20 years old	77	24.44 %
	20–30 years old	145	46.03 %
	31–45 years old	48	16.24 %
	46 years old and above	45	13.29 %
Qualifications	High School Certificate	125	39.64 %
	Bachelor's Student	135	42.91 %
	Postgraduate Student	55	17.45 %
Usage Frequency	Everyday usage	113	36.00 %
	Monthly usage	159	50.55 %
	Usage every six months	30	9.45 %
	Over six months	13	4.00 %
Total		315	100 %

4. Results

4.1. Data analysis

We conducted the statistical analysis with SPSS 29.0 and AMOS 29. Specifically, SPSS was used to analyze the reliability using Cronbach's Alpha and validity through CFA (Brown, 2015), while AMOS was used to examine the relationship between the structural model and the hypotheses shown in Fig. 3 (Anderson and Gerbing, 1988; Shiau et al., 2019).

Reliability and validity of data were confirmed in this study. As shown in Table 3, we used Cronbach's α to confirm reliability. All measures exceeded the recommended threshold, with the lowest Cronbach's α (0.885 for accordance) and CR (0.884 for accordance) values being much greater than 0.7 (Cronbach, 1951; Werts et al., 1974). Then, convergent validity was confirmed based on factor loading values and average variance extracted (AVE) for all constructs and items (Fornell & Larcker, 1981). As Table 3 shows, the factor loadings ranged from 0.819 to 0.914, and the AVE values, all above 0.5, demonstrating convergent validity (Chin, 1998). Furthermore, the variance inflation factor (VIF) values of the subconstruct are below 5, the generally accepted threshold for VIF is 10. If the VIF is below this threshold, multicollinearity is typically not considered a serious issue (Hua et al., 2020).

Discriminant validity substantiates the distinction between a prospective factor and other potential factors in question. In this research, we confirmed discriminant validity using both of Fornell-Larcker criterion and Heterotrait-Monotrait (HTMT) ratio shown in Tables 4 and 5. The Fornell-Larcker criterion, as shown in Table 4, asserts that for each variable, the square root of its AVE exceeds the correlation coefficients with all other variables (Fornell & Larcker, 1981). The results of the HTMT analysis, as presented in Table 5, demonstrate that all HTMT values are below 0.90 (Yusoff et al., 2020). On these facts, reliability and convergent validity were confirmed in this study. Also, there are no discriminant validity issues in this research.

To address potential common method bias (CMB) issues in this study, analytical precedence (Zhang et al., 2025; Podsakoff et al. (2012) was followed: we employed Harman's single-factor test to assess CMB in the sample. According to Harman's single factor rule, common method bias threat does not exist if more than one factor (Podsakoff et al., 2003) has (1) an eigenvalue greater than 1, and (2) the first factor explains less than 50 % of the variance. All factors explored in this study had eigenvalue greater than 1. Also, the variance explained by the first factor was 42.15 %, which is below the critical threshold of 50 % established by Podsakoff et al. (2003). On these results, CMB is not a concern in this

Table 3
Results of internal and convergent reliabilities.

	Item	Factor Loading	Cronbach's α	CR	AVE
Problem-Solving	PS1	0.893	0.918	0.918	0.789
	PS2	0.894			
	PS3	0.878			
Confirmation	CF1	0.892	0.930	0.929	0.815
	CF2	0.914			
	CF3	0.903			
Trust	TS1	0.903	0.898	0.907	0.766
	TS2	0.872			
	TS3	0.819			
Satisfaction	SF1	0.866	0.907	0.899	0.749
	SF2	0.884			
	SF3	0.875			
Perceived Ease of Use	PEU1	0.845	0.885	0.884	0.719
	PEU2	0.840			
	PEU3	0.859			
Continue Using Intention	CUI1	0.885	0.913	0.913	0.777
	CUI2	0.882			
	CUI3	0.877			

Table 4
Results of internal and convergent reliabilities.

Construct	PS	CF	TS	SF	PEU	CUI
PS	0.888					
CF	0.558	0.903				
TS	0.247	0.454	0.875			
SF	0.416	0.472	0.426	0.865		
PEU	0.441	0.468	0.434	0.473	0.848	
CUI	0.409	0.395	0.379	0.464	0.462	0.881
AVE	0.789	0.815	0.766	0.749	0.719	0.777

Note(s): Diagonal elements (bold) are the square root of AVE for each construct. PS: problem-solving, CF: confirmation, TS: trust, SF: satisfaction, PEU: perceived ease of use, CUI: continue using intention

Table 5
HTMT Results.

	PS	CF	TS	SF	PEU	CUI
PS						
CF	0.558					
TS	0.247	0.455				
SF	0.416	0.472	0.426			
PEU	0.441	0.468	0.434	0.472		
CUI	0.409	0.395	0.394	0.464	0.462	

Note(s): PS: problem-solving, CF: confirmation, TS: trust, SF: satisfaction, PEU: perceived ease of use, CUI: continue using intention

study.

Furthermore, the goodness of fit measure based on the following statistical recommendations: χ^2 statistics/degree of freedom (df) should be less than 3 (Bagozzi & Yi, 1988). The comparative fit index (CFI), Tucker-Lewis index (TLI) and incremental fit index (IFI), all of them should be greater than 0.9 (Anderson & Gerbing, 1988; Browne & Cudeck, 1992; Marsh et al, 1988), the root mean square error of approximation (RMSEA) should not exceed 0.05 (Shi et al., 2020). In accordance with the criteria delineated in Table 6, all fit indices met the specified requirements. Consequently, the measurement model demonstrated a commendable level of fit.

4.2. Hypotheses test

In this study, we used SEM to test the model, SEM is a statistical modelling technique emphasizing the examination of relationships among latent variables, well-suited for analyzing models with larger sample sizes and intricate path structures (Bagozzi et al, 1991). Hair et al (2014) advocate the utilization of SEM for identification of pivotal driver constructs, formulation and validation of theories, exploratory research endeavours, and the expansion of prevailing structural frameworks (Anderson and Gerbing, 1988). In this study, we developed a problem-solving ECM research model combined with TAM to understand AI chatbot characteristics in e-commerce platform, and to investigate the intention of customer using chatbot service. Through the integration of variables not previously addressed in extant research and their amalgamation with the ECM, we formulated and empirically examined a novel model for problem-solving capacity of chatbot service, and SEM analysis was carried to examine the structural model and test the hypotheses in this study.

Table 6
Model Fit Testing.

Items	Goodness-of-fit measures	Results
CMIN/df	≤ 3.00	1.508
RMSEA	≤ 0.05	0.040
IFI	≥ 0.90	0.985
TLI	≥ 0.90	0.982
CFI	≥ 0.90	0.985

The directionalities and significance of the hypotheses were assessed using the SEM algorithm. The results of the structural model test are displayed in Fig. 4 and Table 7. Of the twelve hypotheses in the research model, eight were supported. As predicted, problem-solving had a statistically significant positive effect on confirmation ($\beta = 0.567$, $p < 0.001$), confirming H1. Similarly, confirmation positively influenced satisfaction ($\beta = 0.272$, $p < 0.001$), perceived ease of use ($\beta = 0.481$, $p < 0.001$), and trust ($\beta = 0.321$, $p < 0.001$), validating H2, H3, and H4. Additionally, perceived ease of use positively impacted satisfaction ($\beta = 0.260$, $p < 0.001$), trust ($\beta = 0.279$, $p < 0.001$), and the intention to continue using ($\beta = 0.271$, $p < 0.001$), supporting H5, H6, and H7, respectively. Satisfaction also had a significant positive effect on the intention to continue using ($\beta = 0.277$, $p < 0.001$), supporting H8. While trust showed a moderate influence on both satisfaction ($\beta = 0.189$, $p < 0.01$) and the intention to continue using ($\beta = 0.145$, $p < 0.01$), these findings support H9 and H10.

In Table 8, the standardized effects of the variables are further shown. Perceived ease of use (PEU) emerged as the most significant factor, with a total standardized effect of 0.401 (comprising a direct effect of 0.269 and an indirect effect of 0.132). Satisfaction (SF) exhibited the largest direct effect (0.280), while confirmation (CF) had the largest indirect effect (0.335). Conversely, trust (TS) and problem-solving (PS) (both direct and indirect) had minimal influence, indicating that these factors had a relatively smaller impact on users' intention to continue usage.

5. Discussion and conclusions

5.1. Summary of finding

The primary objective of the present study was to understand the motivational factors that contribute to customer's sustained intention to use AI chatbots in e-commerce platforms. This investigation was conducted through the application of an integrated research model, primarily formulated based on the problem-solving, ECM and TAM models taking into consideration also relevant theoretical features in the D&M IS Success model. The specific findings derived from this study can be succinctly summarized as follows:

Considering the first research question (RQ1), "what is the impact of AI chatbot customer service's problem-solving capability on users' sustained usage intention in e-commerce platforms?", this study yields the following findings. The primary focus is on the effect of a specific attribute, namely problem-solving, on consumers' responses. The satisfaction of consumers with the AI chatbot's problem-solving abilities in e-commerce platform is positively correlated with users' intention to continue using the chatbot. Regarding the second question (RQ2), "what is the relationship between perceived ease, based on problem-solving, and consumers' intention to sustain usage in e-commerce platforms?", the study sought to understand the impact of a specific attribute – perceived ease of use. Consumers' perceived ease of use of the AI chatbot's problem-solving in e-commerce platforms is positively associated with users' intention to continue using the chatbot. In the investigation of the third research question (RQ3), which "how does consumer trust influence their responses to the continued usage intention of chatbot customer service in e-commerce platforms?", the study focused on the impact of trust on consumer's responses. The trust that consumers place in the AI chatbot's problem-solving abilities in e-commerce platform is positively correlated with users' intention to continue using the chatbot. However, from the measurement results, it was observed that the outcomes of paths related to trust are relatively small, and the Bias-Corrected Percentile values for the two paths are close to 0. Therefore, this study suggests that the hypothesis for the third research question (RQ3) is only slightly positive towards users' intention to continue usage.

In summary, consumers' perceived ease of use of AI chatbots positively influences their intention to use them. Similarly, satisfaction with AI chatbots based on problem-solving ease of use positively affects their

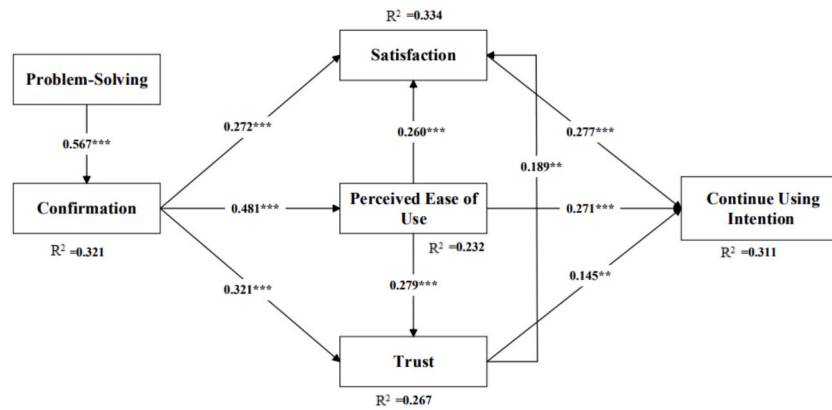


Fig. 4. Results of research model Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 7

Results of the research model.

Hypothesis	Std.	S.E.	t-value	p-value	Significance
H1. Problem Solving → Confirmation	0.567	0.060	10.081	***	Supported
H2. Confirmation → Satisfaction	0.272	0.060	4.183	***	Supported
H3. Confirmation → Ease of Use	0.481	0.050	8.109	***	Supported
H4. Confirmation → Trust	0.321	0.053	4.935	***	Supported
H5. Ease of Use → Satisfaction	0.260	0.073	3.918	***	Supported
H6. Ease of Use → Trust	0.279	0.064	4.196	***	Supported
H7. Ease of Use → Intention	0.271	0.069	3.859	***	Supported
H8. Satisfaction → Intention	0.277	0.059	4.224	***	Supported
H9. Trust → Satisfaction	0.189	0.072	2.965	**	Supported
H10. Trust → Intention	0.145	0.065	2.293	**	Supported

Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8

Effect on the continue using intention.

Factors	Boot SE	Boot LLC	Boot ULC	Effect Direct	Indirect	Total
PS	0.028	0.139	0.251	–	0.190	0.190
CF	0.036	0.265	0.406	–	0.335	0.335
PEU	0.065	0.272	0.524	0.269	0.132	0.401
SF	0.072	0.136	0.417	0.280	–	0.280
TS	0.071	0.014	0.295	0.150	–	0.150

intention to continue usage. Therefore, most consumers perceive that using AI chatbots can effectively address issues encountered while shopping online on e-commerce platforms. However, the process of using AI chatbots is not perceived as particularly convenient. Additionally, the relevance between problem-solving and user trust is notably weak. Our interpretation of this finding suggests that when faced with issues during online shopping, consumers may prioritize the customer service capabilities of AI chatbots to understand and resolve problems over other service methods. Moreover, throughout the usage process, systems that are easy to operate directly or indirectly influence customers' intention to use them.

6. Theoretical contributions

This study makes several theoretical contributions, and some core contributions are summarised in Table 9.

Firstly, it enhances the ECM (Extended Technology Acceptance Model) framework by expanding its application to AI chatbots within

the e-commerce context. Historically, the ECM framework has predominantly focused on the domain of computer, IT and smartphone technology (Ambalov, 2018; Park, 2020; Steelman & Soror, 2017). This extension augments the ECM framework by delving into the problem-solving capabilities of AI chatbots and examining how such capabilities impact consumer satisfaction, trust, and ease of use within the e-commerce platform. Furthermore, this research contributes to the user-perception literature through the development of a comprehensive theoretical model. The model provides insights into the dynamics of user interactions with AI chatbots, shedding light on the factors influencing users' satisfaction, trust, and perceptions ease of use within the e-commerce environment. This contribution advances our understanding of user perceptions in the realm of AI-driven interactions, enriching the existing body of literature on this subject.

The understanding of AI chatbot problem solving abilities is limited and this study offers relevant knowledge development insights. Problem solving is the ability of AI chatbot customer service to accurately understand and solve each customer's problem immediately. As noted by Sadhu et al (2024), the central aim of using AI customer service is to help improve work efficiency and productivity (given information quality benefits – DeLone & McLean, 2003) towards ensuring system quality and user satisfaction (Sadhu et al., 2024; Rane et al., 2023). The findings from this study enhance that notion. This study not only enhances the understanding of the influence of, but also also reinforces the advocacy for more research on problem solving capacities in AI chatbot design and influence on ease of use, satisfaction, trust, and continued usage (Rane et al., 2023; Fu et al., 2020). Effectively designing AI chatbot for effective problem solving would not only drive customers satisfaction but also boost continued usage intention (Rane et al., 2023).

This study addresses the fundamental requirements of customer service in e-commerce through the lens of problem-solving (Chumpitaz & Paparoidamis, 2020). Differing from Park (2020), this research considers both problem-solving and perceived ease of use as critical boundary conditions influencing consumers' willingness to use in this context. While prior research on customer responses to AI chatbots has primarily concentrated on attributes such as anthropomorphism (Hsiao and Chen, 2022; Lu et al., 2019) and user interface (Hsiao and Chen, 2022; Wang et al., 2019; Przegalinska et al., 2019), the identified boundary conditions fall short of adequately showcasing the problem-solving capabilities. Consequently, even in scenarios where consumers exhibit a lack of trust in AI chatbots, the proficiency in problem-solving still exerts a significant influence on consumers' inclination to persist in using the service. Therefore, the recognition of function-based boundary conditions emerges as a pivotal avenue for future research.

A closer look at the directional influence of constructs explored in this study offers insights that further enhance domain literature. This study illuminates the direct and indirect impact of perceived ease of use on consumers continued usage intention. Illuminating these aspects

Table 9

A Summary of core contributions.

Nr.	Construct/ Context	Relevant Prior Literature	Contribution Summarised
1	Ease of Use	Petter et al. (2008), DeLone & McLean (2003)	This study illuminates the direct and indirect impact of this construct on consumers continued usage intention. Illuminating these aspects enhance theoretical notions forwarded by prior literature (elaborated in the body of text below). In this study, satisfaction exhibited the largest direct effect. Taken along with the earlier point on the direct and indirect effect of perceived ease of use, this study further enhances not only the satisfaction – consumers continued usage intention association but also the potential perceived ease of use connection to that association (elaborated further in the body of text)
	Satisfaction		
2	Confirmation	Cheah et al. (2023), Almogren & Aljammaz (2022), Aldammagh et al. (2021), Alamsyah et al. (2022), Park (2020)	In this study, a significant impact of confirmation on ease of use, user satisfaction and trust respectively were found. Also, this study suggests that problem solving significantly determined confirmation. These insights enhance domain literature (elaborated in the body of text).
3	Trust	Manzoor et al. (2020), Przegalinska et al. (2019), Nguyen et al. (2021), Cheng et al. (2021)	The insights about the impact of trust on continued usage intention and on user satisfaction reinforce domain literature. Unlike perceived ease of use (PEU), trust showed a lesser impact on customers continued usage intention. However, perceived ease of use has a significant direct impact on trust. Trust, however, also significantly impacts user satisfaction. Combining the insights relating to the impact of trust in tandem with the insights on ease of use, domain literature is also enhanced.
4	Problem Solving Capabilities in AI Chatbot	Myin (2024), Sadhu et al (2024), Rane et al. (2023) Park (2020) Hsiao and Chen, 2022; Lu et al., 2019; Wang et al., 2019; Przegalinska et al., 2019	Taken together, the insights from this study not only enhance the literature but also reinforce the advocacy for more research on problem solving capacities in AI chatbot design and influence on ease of use, satisfaction, trust, and continued usage This study enhances Park (2020): this research considers both problem-solving and perceived ease of use as critical boundary conditions influencing consumers' willingness to use. Unlike prior research on

Table 9 (continued)

Nr.	Construct/ Context	Relevant Prior Literature	Contribution Summarised
			customer responses to AI chatbots that concentrated on attributes such as anthropomorphism (Hsiao & Chen, 2022; Lu et al., 2019) and user interface (Hsiao & Chen, 2022; Wang et al., 2019; Przegalinska et al., 2019), this current study explored problem-solving capabilities

enhance theoretical notions forwarded by prior literature (Petter et al., 2008; DeLone & McLean, 2003). Demonstrating the importance of the 'Use' construct, DeLone and McLean (2003) commented: "Use must precede "user satisfaction" in a process sense, but positive experience with "use" will lead to greater "user satisfaction" in a causal sense" (cited in Petter et al., 2008, p.238). Further, they forward that increased user satisfaction will intensify intention to use, which also subsequently affects use. Equally illuminated in this current study is the directional influence of user satisfaction on customers continued usage intention. As a matter of fact, satisfaction with AI chatbots as one of the most significant influencing factors of consumers continued usage intention. While that finding resonates with Melián-González et al (2021) that found a highly significant influence of performance of AI chatbots on consumers future usage intention, the additional insights about the directional impact, combined with the perceived ease of use directional impact (direct and indirect), connect reasonably with prior literature explaining the 'Use' construct (Petter et al., 2008; DeLone & McLean, 2003). The findings here also lend support to Nguyen et al.'s (2021) study that found perceived usefulness and the major driver of continued usage intention in the banking sector.

The plausibility of the above Ease of Use – User Satisfaction – Continued Usage Intention conclusion is further supported when the findings for Trust are examined. Supporting the conceptualisation, Trust has a significant impact on continued usage intention as well as on user satisfaction, insights that lend support to prior literature (Przegalinska et al., 2019; Nguyen et al., 2021; Cheng et al., 2021). Examining the trust findings in tandem with earlier conclusion (Ease of Use – User Satisfaction – Continued Usage Intention link) this study also extends aforementioned literature: though significant, Trust is a lesser determinant of Continued Usage Intention, compared to Ease of Use, given that trust impacts User Satisfaction, and Ease of Use impacts significantly both Continued Usage Intention (direct and indirect) and User satisfaction, there is a plausibility to conclude that the overall impact of trust may be mediated by the influence of Ease of Use (Ease of Use impacts significantly on Trust).

According to TAM theory, perceptual and utilitarian factors of users' use hinge significantly on a combination of the product itself, related services and users' actual feelings during use (Cheah et al., 2023; Almogren & Aljammaz, 2022; Aldammagh et al., 2021). On that foundation, validation (confirmation) concept has often been viewed as a determinant of perceived usefulness and perceived ease of use (Alamsyah et al., 2022; Park, 2020). Park (2020) suggests: (1) based on 1380 respondents that users' confirmation of smart wearable devices plays a significant role in perceived ease of use, usefulness, and enjoyment; and (2) based on 363 respondents that perceived usefulness, attitude, concentration, and perceived behavioural control significantly determined their confirmation. This current study offers insights that enhance confirmation dynamics: confirmation has a significant positive impact on ease of use, user satisfaction and trust, while problem solving impacts significantly on confirmation.

6.1. Practical implications

Our research also has several practical implications for both AI technology developers and companies providing customer service. Firstly, the use of AI-based chatbots on e-commerce platforms is already widespread. However, the predominant focus of most AI chatbots is on product category recommendation (Zhou et al., 2023; An et al., 2023; Wu, 2019). While AI chatbots contribute to businesses in marketing their products during online consumer shopping, online stores should also be attentive to consumers' resistance intentions (Cheng et al., 2021). The findings of this research can capture the attention of online stores and technology developers who are oriented towards the problem-solving functionality of AI chatbots. This orientation is pivotal for online retailers seeking AI chatbots that are not only more effective in addressing customer issues than human customer service but also more efficient and cost-effective.

Furthermore, the study emphasizes the paramount role of cultivating consumer trust in chatbot capabilities to ensure customer retention. Establishing confidence in the chatbot's ability to fulfil its intended purpose is critical for fostering long-term relationships between consumers and e-commerce platforms. The research also suggests a noteworthy distinction in the factors influencing continued chatbot usage intentions. While perceived usefulness emerges as a more significant determinant, surpassing the impact of ease of use, developers should focus on enhancing the practical value that chatbots provide to users to encourage sustained engagement. Additionally, the study underscores the strategic value of managing consumer expectations and confirming chatbot success. E-commerce providers should actively communicate and align user expectations with the capabilities of chatbots, ensuring a positive user experience and mitigating any potential disappointment. User satisfaction is a critical determinant of customer continued usage intention; therefore, system design efforts should be geared towards ensuring user satisfaction. Furthermore, given the culminating impact of 'use' (perceived ease of use in this study) (Petter et al., 2008; De Lone & McLean, 2003), systems designers should also give due consideration to perceived ease of use, given the directional (direct and indirect) impact on continued usage intention and user satisfaction impact (in-between). The significant influence of perceived ease of use on trust should also be noted; however, trust has lesser impact on continued usage intention.

Considering these findings, the study recommends that e-commerce providers invest in comprehensive chatbot training programs. This investment is seen as instrumental in boosting the utility and accuracy of chatbots, enhancing their effectiveness in resolving customer issues. Overall, the study's insights offer valuable guidance for the strategic development and deployment of chatbots in the e-commerce landscape.

6.2. Limitations and future research

This research has several limitations. First, while the study acknowledges the impact of various attributes on consumers' intentions to continue using chatbots, it primarily focuses on addressing specific problems. Future research should conduct a more detailed examination of the distinct capabilities of AI chatbots. Second, it is important to note that this research exclusively targets chatbots within e-commerce platforms. Unlike chatbots used in healthcare, digital economy, and education, the findings may have limited generalizability. Third, although the total sample size of 315 usable responses is adequate for preliminary analysis, it falls short of the ideal sample size of 500 recommended for SEM techniques, as outlined by Kyriazos (2018). This limitation in sample size can adversely affect the statistical estimation properties, such as standard errors and statistical power (Wolf et al., 2013). Increasing the sample size to 500 or more participants would enhance the robustness and confidence in the study's conclusions.

Methodologically, only Chinese AI chatbots platform was considered in this study. Given heterogeneity in national culture (Opute, 2015; Hofstede, 1980) and the significant influence of culture on consumers'

behaviour (Opute et al., 2022a, 2022b; Opute, 2017), further studies that consider this model in other settings would enhance general understanding of this topic. Further on the point of research framework, future research could address the limitations of this research in this regard by exploring frameworks beyond the premise in this research. A core direction in that path could be to explore the levels of mediation that exist between the contextualised variables. For example, future research could illuminate the mediation effect of perceived ease of use, satisfaction, and trust on the relationship between problem-solving confirmation and continued usage intention. Future research could also consider how customer centric emotions (e.g., angry vs. calm) moderate these relationships. Future research should not only consider the attributes of chatbots but also delve into their capabilities. A further limitation of this study relates to the fact that it focused on the general perceptions rather than experimental manipulation. Future research could also address this limitation. Addressing these limitations and potentially expanding the investigation beyond the e-commerce context would provide a more comprehensive understanding of AI chatbots.

CRediT authorship contribution statement

Junzhe Gao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Abdullah Promise Opute:** Writing – original draft, Validation, Software, Data curation, Conceptualization. **Caroline Jawad:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Meng Zhan:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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